

29 Calculating total resistance across XY, R_{XY} : $\frac{1}{R_{XY}} = \frac{1}{4.0 + 4.0} + \frac{1}{8.0} \Rightarrow R_{XY} = 4.0 \Omega$

Calculating total resistance of whole network of resistors, R_T :

$$\frac{1}{R_T} = \frac{1}{4 + R_{XY}} + \frac{1}{8} \Rightarrow R_T = 4.0 \Omega$$

Using potential divider rule, terminal p.d. $V = \frac{R_T}{R_T + 6.0} \times 20 \text{ V} = \frac{4.0}{4.0 + 6.0} \times 20 \text{ V} = 8.0 \text{ V}$

Using potential divider rule, $V_{XY} = \frac{R_{XY}}{R_{XY} + 4} \times 8.0 \text{ V} = \frac{4.0}{4.0 + 4.0} \times 8.0 \text{ V} = 4.0 \text{ V}$

Answer: B

20 Using Fleming's Right Hand Rule, the direction of the induced current in the disc can be